

Laboratory 2

TASK: MULTIPROCESSING WITH QUEUE AND MUTEX
(use **python** language and **Windows OS**)

1. Use the example run.py, create a parallel version using multiprocessing.Process
 - Structure the program into producer/consumer or worker roles as needed.
 - Ensure multiple processes can run safely in parallel. (1 point)
2. Implement a multiprocessing Queue structure for inter-process efficient data exchange.
 - Show enqueue/dequeue operations.
 - Demonstrate safe transfer of multiple data items. (1.5 points)
3. Use Mutex Lock to manage concurrency and avoid race conditions.
 - Apply locks during shared resource access.
 - Show where locking is required and why. (1 point)
4. Implement proper process communication and orchestration.
 - Ensure processes terminate safely.
 - Use join() to synchronize. (0.5 point)
5. Add exception handling and timeouts.
 - Use AsyncResult.get(timeout=...) or similar pattern.
 - Catch and handle multiprocessing.TimeoutError. (0.5 point)
6. Confirm correct results with the sequential version and compare the running time (use time.time() function) (0.5 point)

NOTE: Upload the print-screen of final code execution on enauczanie and of course your code in python :)